

Table 1

Observed mean home range size and mean proportion of exclusive area $\pm$ standard error in founder and field-born resident females

	Founder females	Field-born females	
		Reprod	Nonreprod
50% Kernel home range	48.96 $\pm$ 2.84	43.69 $\pm$ 5.80	36.45 $\pm$ 3.87
Proportion exclusive area	0.70 $\pm$ 0.03	0.46 $\pm$ 0.08	0.37 $\pm$ 0.07

The values for field-born females are separated into reproducing (reprod) and nonreproducing females (nonreprod).

rather than dispersers. The total mean proportion of juveniles becoming residents was 0.19 ± 0.07 standard error in summer and 0.33 ± 0.15 in winter. Additionally, reproduction in the increasing populations stayed high well into winter, and several juvenile females started to reproduce as well (mean proportion of field-born females reproducing in summer: 0.53 ± 0.23 and in winter: 0.18 ± 0.08). The proportion of adult females in the populations correlated negatively with the number of resident females throughout the seasons (Figure 2c). In the 3 decreasing populations, all females were reproducing. In the increasing populations, however, the fraction of adult females decreased toward the winter because nonreproducing females were added to the populations due to reproduction.

Home range size and exclusiveness

The observed home range sizes and proportion of the ranges used exclusively by females can be found in Table 1. We selected the best model by applying a log-likelihood ratio test at each step of the selection procedure. Breeding status did not enter the best models of home range size and exclusive use of home ranges, and is not reported. However, effects of season and the proportion of reproducing females will be reported because of their importance. The model describing variation in home range size showed that older females had larger home ranges than younger females, and season and the proportion of reproducing females in the population did not affect home range size (Table 2).

The best model explaining area of exclusive use showed lower exclusive use in winter than in summer and a negative effect of female density on exclusive use in adult females, whereas young females did not respond to density (Table 2, Figure 3). The fraction of reproducing females in the populations did not affect proportion of exclusive use of the home range area (Table 2).

DISCUSSION

Home range size of female root voles was larger for founder individuals than for field-born individuals, and there was no effect of season or density on home range size. All females used a lesser proportion of their home range exclusively in winter than in summer. Additionally, founder females decreased exclusiveness with increasing density of resident females, whereas field-born females did not show any response in exclusive use of home range with density. At high density, therefore, the difference in exclusiveness between age classes disappeared irrespectively of season (Figure 3).

A gradual increase in density from spring to fall and early winter is known from most small rodents that fluctuate in numbers throughout the year (Erlinge et al. 1990; Turchin and Ostfeld 1997; Crespin et al. 2002; Priotto et al. 2002; Lima et al. 2003). However, because we minimized the increase in

Table 2

Model estimates with standard error, *F* values, and probability for home range size and the proportion exclusive use of home ranges (exclusiveness)

Model	Estimate $\pm$ SE	<i>F</i> value	<i>P</i> value
Home range size			
Intercept	3.738 $\pm$ 0.194	$F_{1,123} = 6733.722$	$P < 0.001$
Age class	-0.164 ± 0.105	$F_{1,41} = 5.168$	$P = 0.028$
Prop reprod	0.031 ± 0.198	$F_{1,123} = 0.529$	$P = 0.468$
Season	-0.130 ± 0.103	$F_{1,123} = 1.589$	$P = 0.210$
Exclusiveness			
Intercept	0.118 $\pm$ 0.017	$F_{1,121} = 714.186$	$P < 0.001$
Season	-0.023 ± 0.006	$F_{1,121} = 34.217$	$P < 0.001$
Density	-0.008 ± 0.002	$F_{1,121} = 18.645$	$P < 0.001$
Age class	-0.069 ± 0.016	$F_{1,41} = 5.249$	$P = 0.027$
Prop reprod	-0.001 ± 0.014	$F_{1,121} = 0.008$	$P = 0.930$
Density $\times$ age class	0.010 ± 0.003	$F_{1,121} = 13.604$	$P < 0.001$

The intercept represents estimates for founder females in the summer season, age class represents the difference between founder and field-born females, and season represents the change from summer to winter. Prop reprod, the proportion of females being reproductively active in the population.

density by connecting the source populations with sinks and allowed density to develop freely among our study populations, the population trajectories showed great variation where the densest population consisted of about 70 resident females per hectare and the least dense population had about 7 resident females per hectare (Figure 2a). Thus, we were able to disentangle the effect of season and density.

The seasonal change in the exclusive use of home ranges observed in this study apparently fits with the predictions of the hypothesis that space use is determined by the reproductive condition of females (Erlinge et al. 1990; Puseenius and Viitala 1993; Priotto and Steinmann 1999; Priotto et al. 2002). However, we found no difference in exclusive use between breeding and nonbreeding females, which does not support this hypothesis. We did find a difference in exclusiveness depending on age class, and other studies of reproduction and maturing of females in microtines show that late-born females often do not mature or start breeding the season of

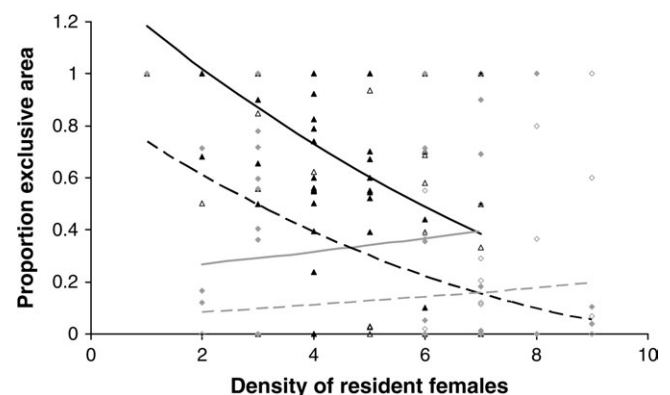


Figure 3

Density and season-dependent variation in the proportion exclusive use of home ranges. Data shown are model predictions (lines) and observed values (points) in summer (solid lines and triangles) and winter (dotted lines and diamonds). Founder females are represented by black lines and solid points, whereas field-born females are represented by gray lines and open points. Data were analyzed on an arcsinus scale for home range exclusiveness.